

Campus asset checklist

Project RescueSwarm · NIDAR 2026–27 · what to confirm before submitting the funding request

Why. Find out what the institute already owns *before* asking it for money. Every confirmed *yes* either reduces the request or removes a risk, and none of it costs the project any capability — it is the same equipment, borrowed instead of bought.

How. Grouped by *where you go*, so each visit is one conversation. **Ask the question as written.** The general question invites a misleading answer: “*do you have a GPS?*” gets a yes that turns out to be a total station. Record the answer, not just yes/no — “yes, but booked until March” is a no.

Section	Value	Effect if confirmed
A · Budgeted equipment	Rs. 70,500	Leaves the request. Ask falls to about Rs. 7,43,000
B · Stock and consumables	~Rs. 15,000	Partial reductions
C · Questions that remove risk	Rs. 25,000+	Resolves the insurance blocker
D · Already assumed	Rs. 3,54,000	Confirm, or the ask goes UP

Section D matters as much as section A. The budget already assumes Rs. 3,54,000 of institute equipment is available to us. If it is not, that money comes back into the request. Confirming it protects the number you are submitting.

A · Equipment that would leave the budget — Rs. 70,500

→ Civil Engineering, survey laboratory

☐ **RTK base receiver** Rs. 18,000 · biggest single item, least likely to just work

*“Do you have a survey-grade GNSS receiver that can run as a **base station** and output **RTCM3 corrections** over serial or IP to a third-party rover, at about 1 Hz? Would it be free for roughly eight field days?”*

Not a match if: it is a total station (not GNSS); it only post-processes and cannot stream live corrections; it is booked for coursework in our window.

Answer: _____

☐ **Survey tripod + tribrach** Rs. 4,000

“A standard tripod with a 5/8-inch thread and a tribrach we could borrow for field days?”

Answer: _____

→ Robotics / aeromodelling club

☐ **Safety-pilot transmitter** Rs. 8,000

*“Do you have an **ExpressLRS** transmitter, or a radio with a **module bay** that can take an **ELRS** module?”*

Not a match if: it is FlySky, Spektrum or any non-ELRS set — wrong protocol, cannot bind to our receivers. A module bay alone is enough; the module is cheap.

Answer: _____

☐ **Battery chargers** Rs. 14,000

*“A charger that balance-charges **6S lithium** at **5 A or more**, and measures **per-cell internal resistance**?”*

Not a match if: it cannot measure internal resistance — that capability is why the separate cell tester was already deleted from the budget.

Answer: _____

→ Mechanical Engineering laboratory

☐ **Calibrated scale** Rs. 12,000

*“A calibrated balance reading to about 1 g up to 10 kg — and **can it leave the building** for a competition day?”*

Not a match if: it is bench-bound. This decides the Rule C2 weigh-in, so it has to travel. If it cannot, we still buy one.

Answer: _____

☐ **Load cell and DAQ** would offset the Rs. 8,000 thrust stand

*“Do you have a **load cell up to about 10 kg with a data-logger or DAQ** we could use for propeller thrust measurement?”*

This one is **load-bearing for the project**: our motors publish no thrust curve, so this measurement decides whether the aircraft hovers.

Answer: _____

→ Departmental stores

☐ **Equipment cases** Rs. 9,000 · foam-lined, must survive transport to the venue

Answer: _____

☐ **Charger PSU** Rs. 1,500 · 12 V at 20 A or better, usable in the field

Answer: _____

→ Institute safety office

☐ **Fire extinguisher, sand, lithium charging bags** Rs. 4,000

*“A **CO₂ or Class D** extinguisher and lithium charging bags that can be **signed out to a field site**?”*

54 lithium cells are cycled across this programme. Thermal runaway is a real failure mode, so this capability is kept whether or not it is borrowed.

Answer: _____

B · Stock and consumables — up to about Rs. 15,000

Drawn from the Rs. 13,000 fasteners line and the Rs. 21,400 stock line. Partial answers still help.

☐ 3D printer filament

the institute already provides the printer

*“Does the department supply **PETG or ASA filament**, or do we buy our own?”*

Answer: _____

☐ Carbon-fibre offcuts

machine shop

*“Do you hold **CF plate or tube stock**, or offcuts from previous projects we could use for spares?”*

Answer: _____

☐ Fasteners

stores

*“Is there a stock of **M3 socket-cap, nyloc and washers** we can draw against?”*

Answer: _____

C · Questions that remove a risk rather than a cost

→ Dean R&D / accounts office

☐ Insurance

resolves a Rs. 25,000 open item *and* a flight blocker

*“Does the institute hold **third-party liability cover that extends to student projects operating unmanned aircraft**? If not, is it required before we fly?”*

This is the single unresolved item in the budget. Cover is commonly mandatory for unmanned operations in India. If the institute already holds a policy that extends to us, the Rs. 25,000 stays out and the flight blocker clears. If it does not, we must know **before any flight**, not after.

Answer: _____

☐ Travel and participation fund

money that is not in this budget at all

*“Is there a **separate fund for competition travel, accommodation and entry fees**, distinct from equipment funding?”*

Our request covers **equipment only**. Getting three aircraft and a team to the venue is costed nowhere. Many institutes fund this from a different head, so it need not compete with the equipment request.

Answer: _____

☐ Sanction thresholds and budget heads

*“What are the **sanction limits**, and which **head** should capital equipment versus consumables be indented under?”*

We need this to submit correctly, not to structure around it. Our tranches are tied to technical milestones and must not be presented as a split indent.

Answer: _____

☐ Flight-testing site

*“Is there an approved **open area on or near campus** we may use for UAV flight testing, and what permission is needed?”*

Answer: _____

D · Confirm what the budget already assumes — Rs. 3,54,000

These are already counted as institutional contribution and *removed* from the request. If any is not actually available, that money comes back into the ask.

	Item	Assumed	Confirmed available?
☐	Ground-station laptop	Rs. 85,000	_____
☐	Training GPUs (model training)	Rs. 75,000	_____
☐	Backup ground-station laptop	Rs. 55,000	_____
☐	Portable power station / field power	Rs. 55,000	_____
☐	3D printer	Rs. 50,000	_____
☐	Soldering station	Rs. 9,000	_____
☐	Rotary tool + CF extraction	Rs. 8,500	_____
☐	Bench power supply	Rs. 8,500	_____
☐	Multimeters	Rs. 8,000	_____
	Total assumed	Rs. 3,54,000	

For the GPUs the question that matters is not “*do you have GPUs*” but:

“*Can we get **scheduled access** for model training, and roughly how many GPU-hours?*”

After the walk-around

1. Total the confirmed items in section A.
2. Regenerate the workbook with those lines zeroed, then re-check the covering note against tab 00 — the ask, the capital share and the institutional contribution all move together.
3. If anything in **D** came back unavailable, add it back to the request *before* submitting.
4. Open the covering note with what you found. “We audited the institute’s existing assets and reduced this request by Rs. X before submitting it” is a materially stronger opening than the same number without it.